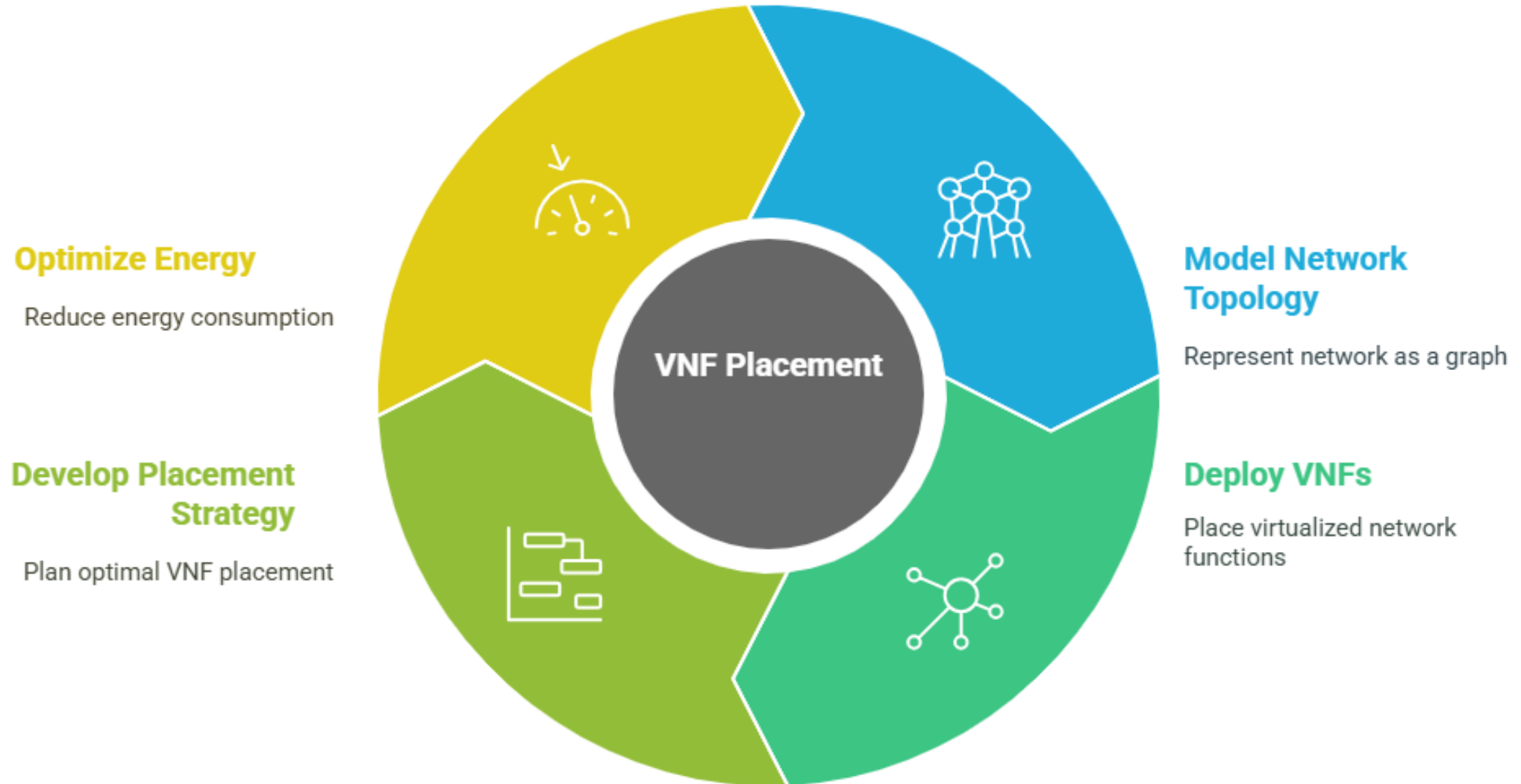




ML-BASED SLICE PLACEMENT

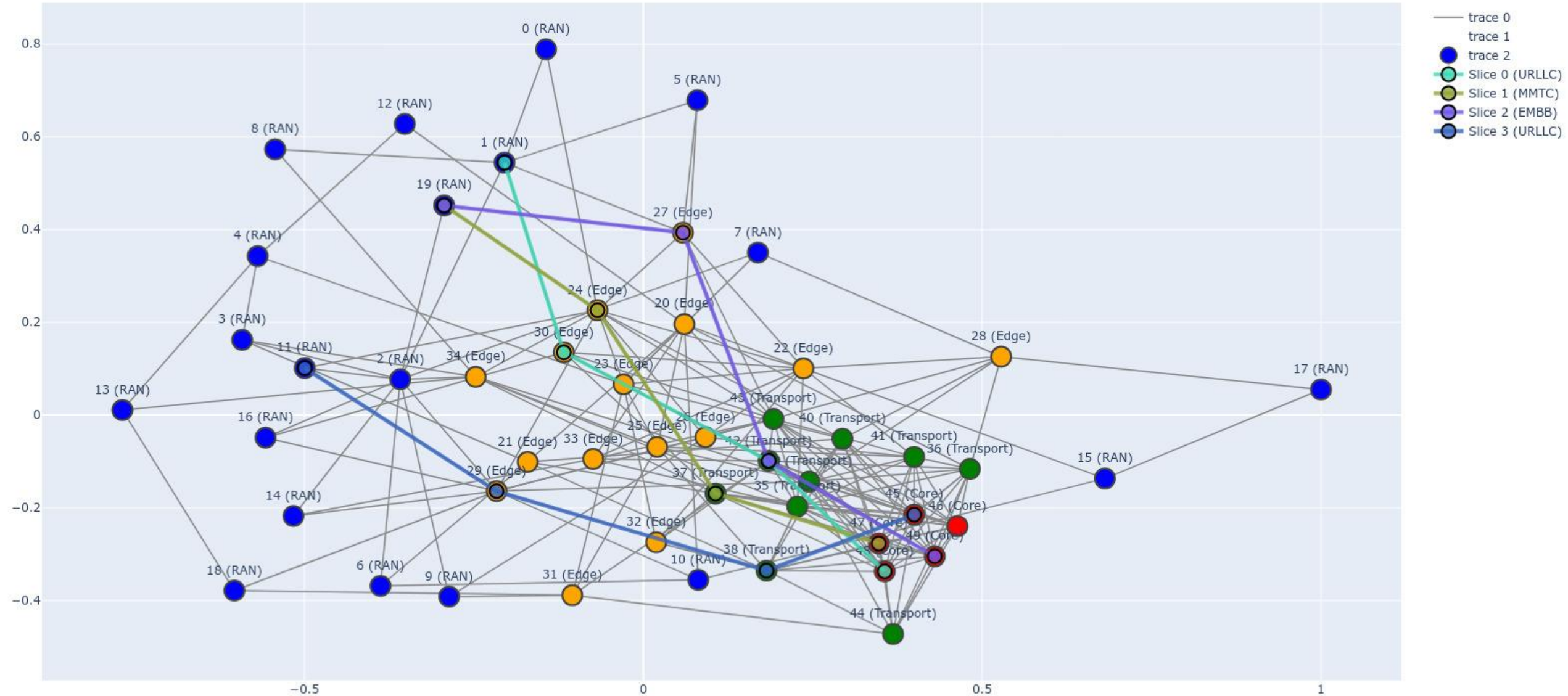
OPTIMIZING ENERGY USAGE

VNF Placement Optimization

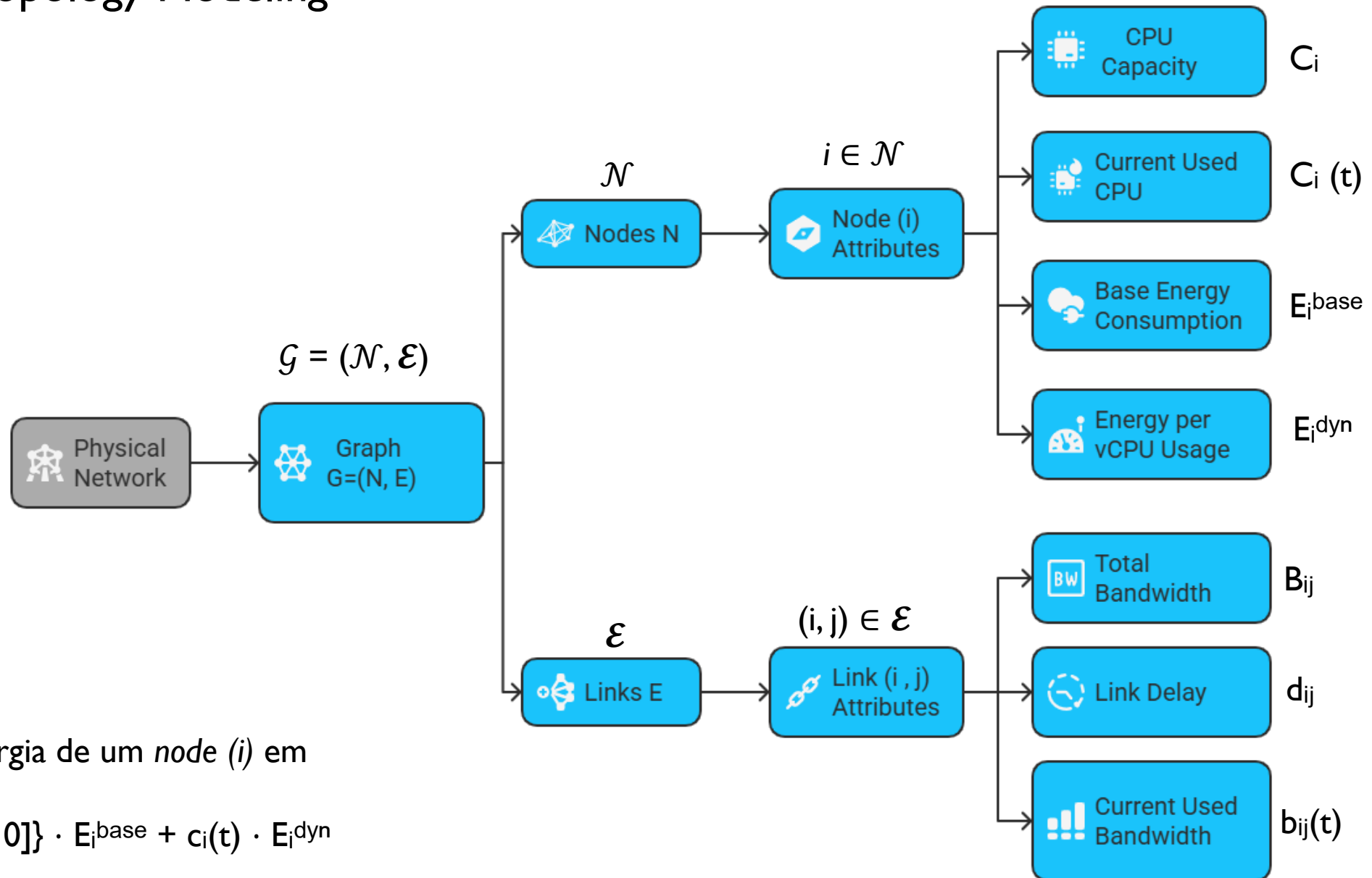


Slice Placement Illustration

Slice Placement Animation - Slice 3 (URLLC) - Energy Usage: 11978.225472711001



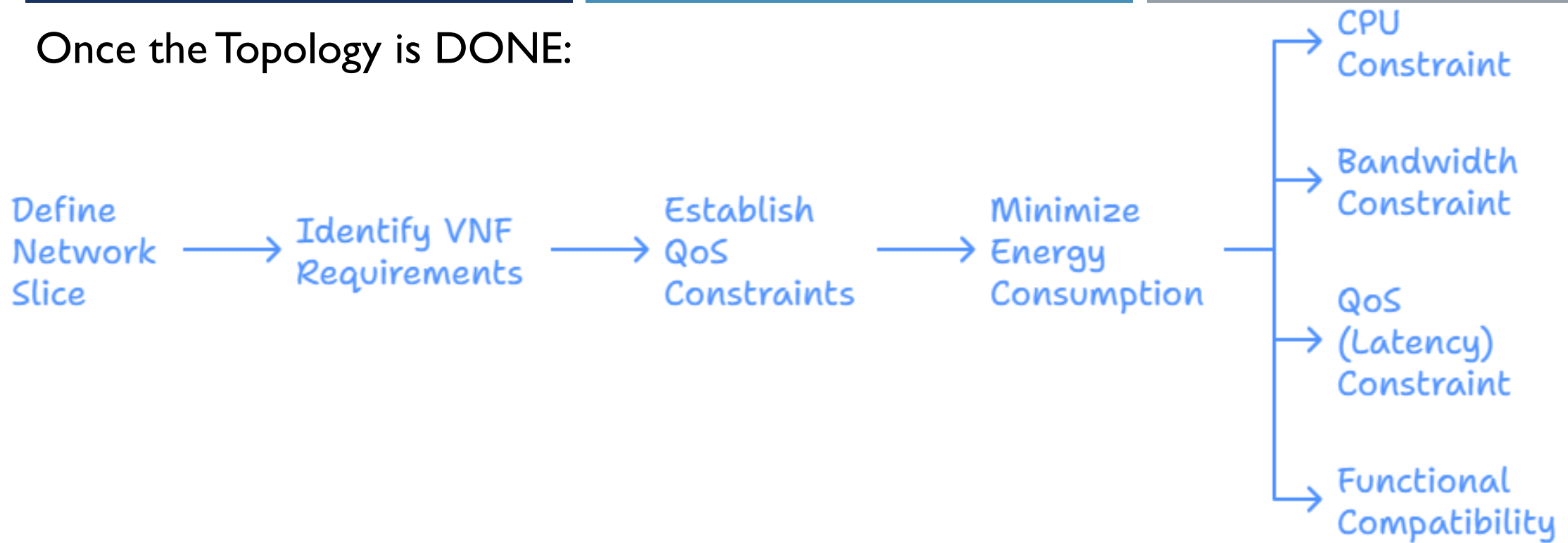
Network Topology Modeling



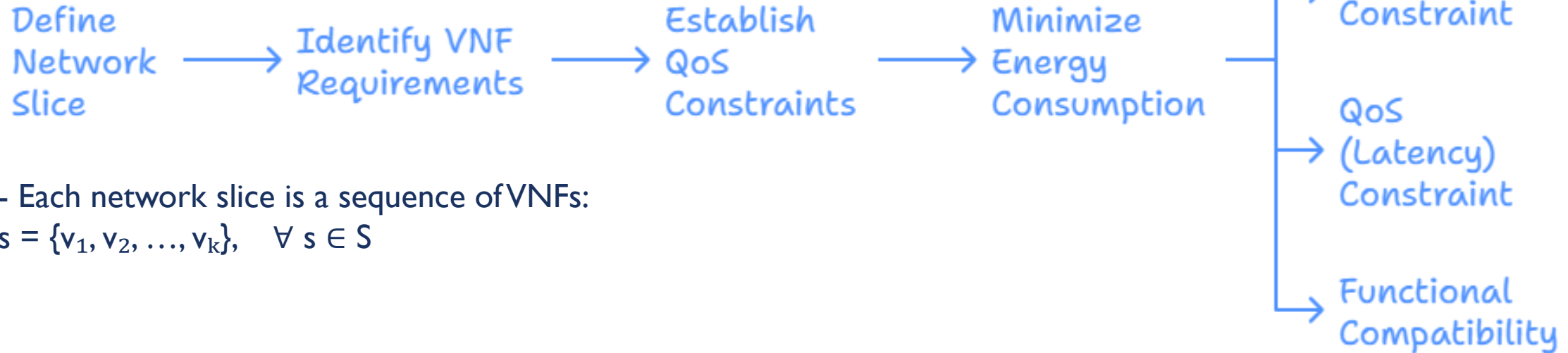
Consumo de energia de um *node* (*i*) em um tempo *t*:

$$E_i(t) = \mathbb{1}_{\{[c_i(t) > 0]\}} \cdot E_i^{base} + c_i(t) \cdot E_i^{dyn}$$

Once the Topology is DONE:



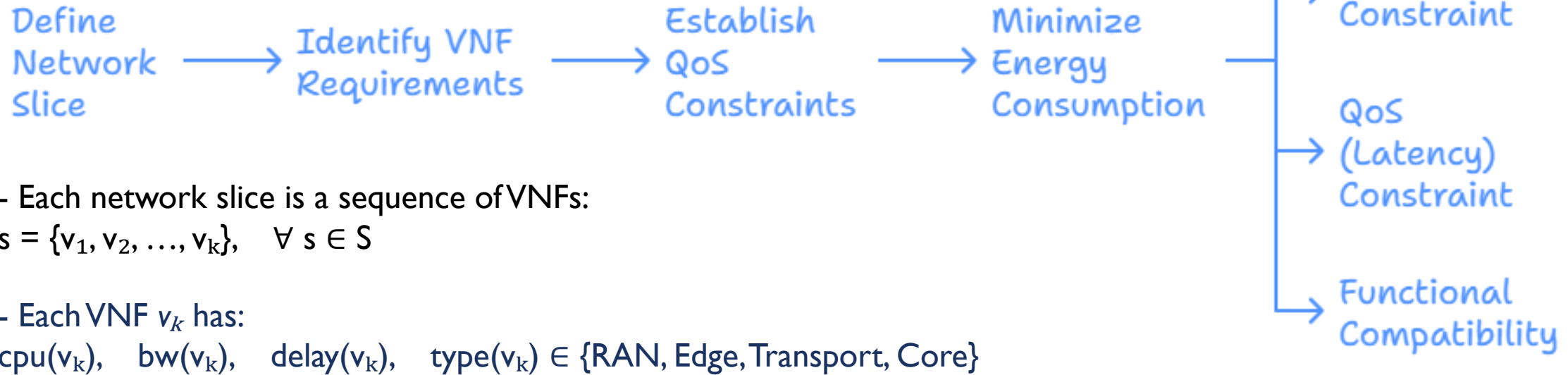
Once the Topology is DONE:



- Each network slice is a sequence of VNFs:

$$s = \{v_1, v_2, \dots, v_k\}, \quad \forall s \in S$$

Once the Topology is DONE:



Once the Topology is DONE:



- Each network slice is a sequence of VNFs:

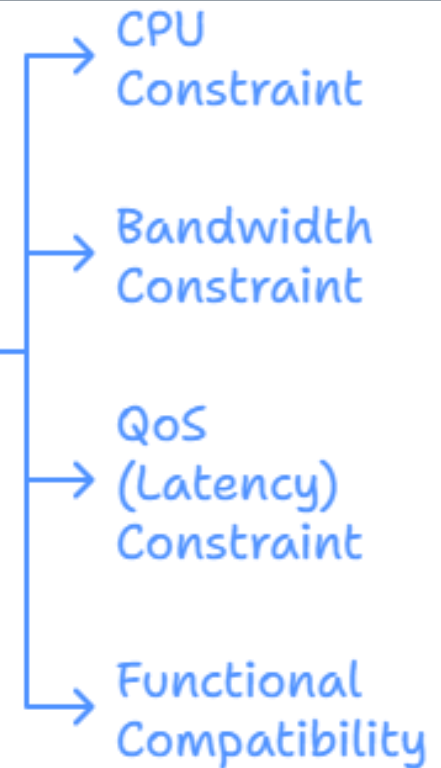
$$s = \{v_1, v_2, \dots, v_k\}, \quad \forall s \in S$$

- Each VNF v_k has:

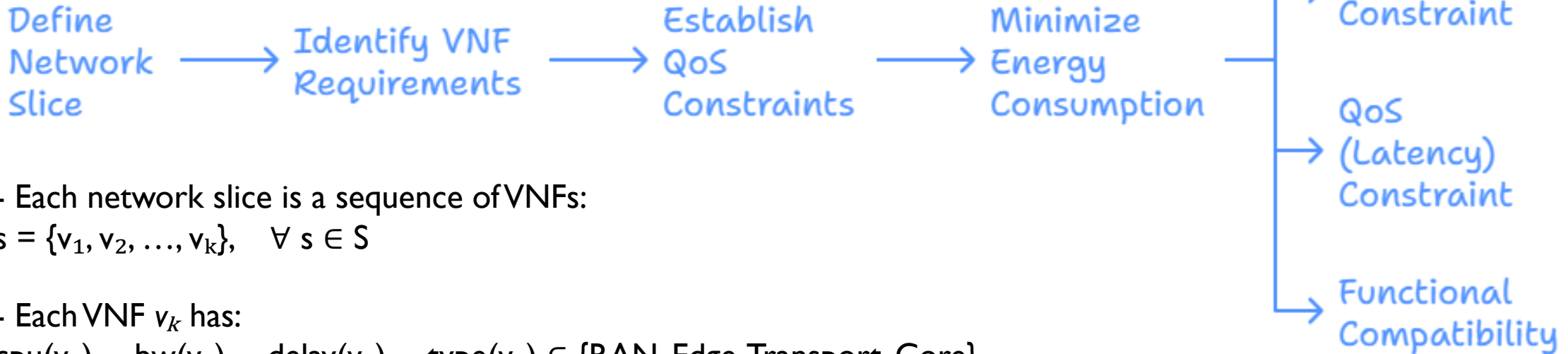
$$\text{cpu}(v_k), \quad \text{bw}(v_k), \quad \text{delay}(v_k), \quad \text{type}(v_k) \in \{\text{RAN, Edge, Transport, Core}\}$$

- Each slice s has QoS Constraints

$$\Delta_s = \text{max latency} \quad \beta_s = \text{min bandwidth} \quad \delta_s = \text{max edge latency}$$



Once the Topology is DONE:



- Each network slice is a sequence of VNFs:

$$s = \{v_1, v_2, \dots, v_k\}, \quad \forall s \in S$$

- Each VNF v_k has:

$$\text{cpu}(v_k), \text{bw}(v_k), \text{delay}(v_k), \text{type}(v_k) \in \{\text{RAN}, \text{Edge}, \text{Transport}, \text{Core}\}$$

- Each slice s has QoS Constraints

$$\Delta_s = \text{max latency} \quad \beta_s = \text{min bandwidth} \quad \delta_s = \text{max edge latency}$$

Therefore, we define the following objective:

$$\min \sum_n [1\{c_n(t) > 0\} \cdot E_n^{\text{base}} + c_n(t) \cdot E_n^{\text{dyn}}]$$

Subjected to:

$$\sum \{v_k \in s: \pi(v_k) = n\} \text{cpu}(v_k) \leq C_n \quad \forall n \in N$$

$$\sum \{(v_k, v_{k+1}) \in s\} \text{bw}(v_k) \leq B_{\{\pi(v_k), \pi(v_{k+1})\}}$$

$$\sum \{(v_k, v_{k+1}) \in s\} d_{\{\pi(v_k), \pi(v_{k+1})\}} + \sum \text{delay}(v_k) \leq \Delta_s$$

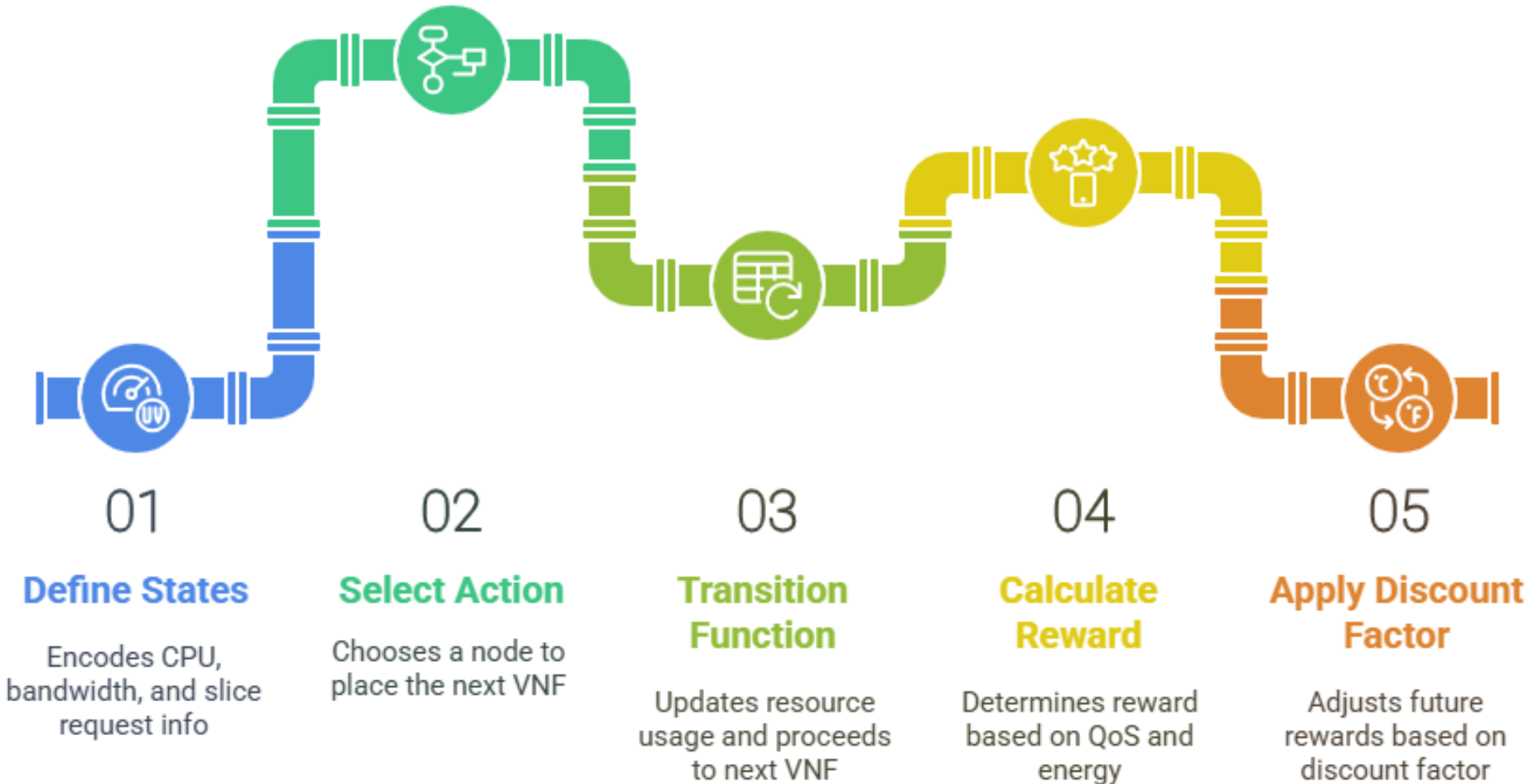
- CPU Resources

- Bandwidth Availability

- Latency Boundaries

VNF PLACEMENT – APPROACH PROPOSAL

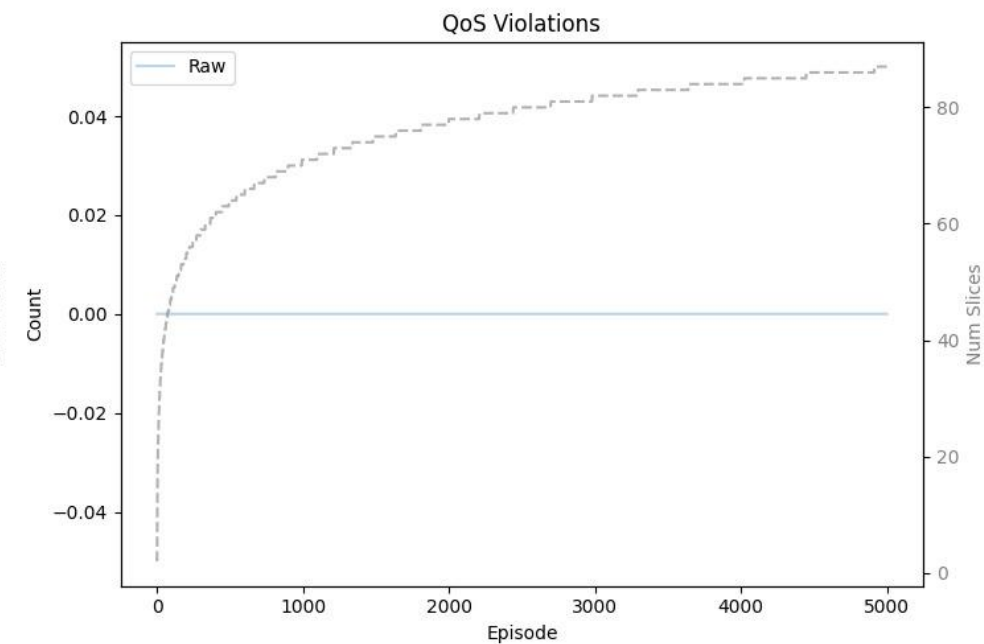
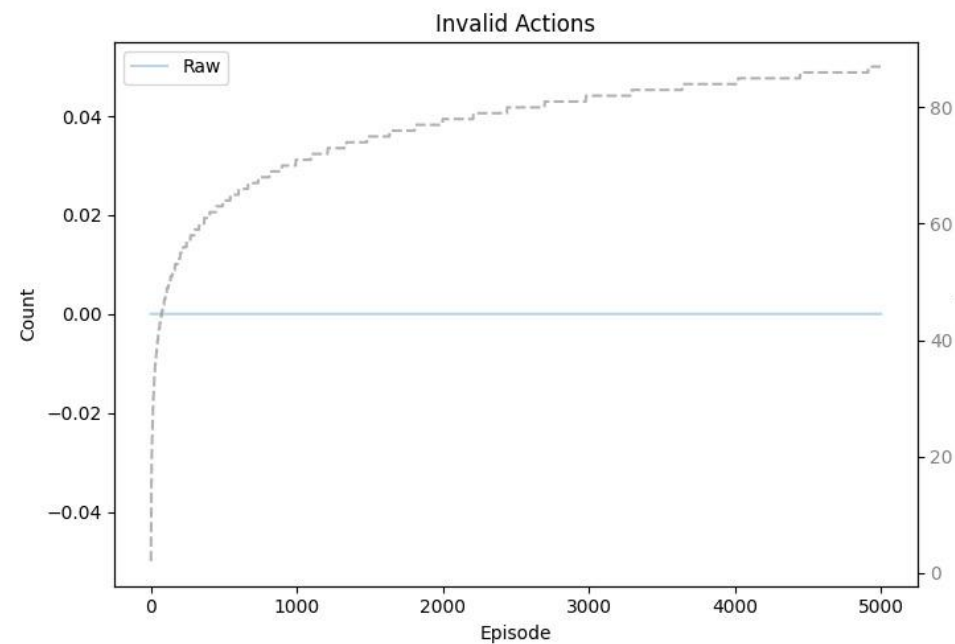
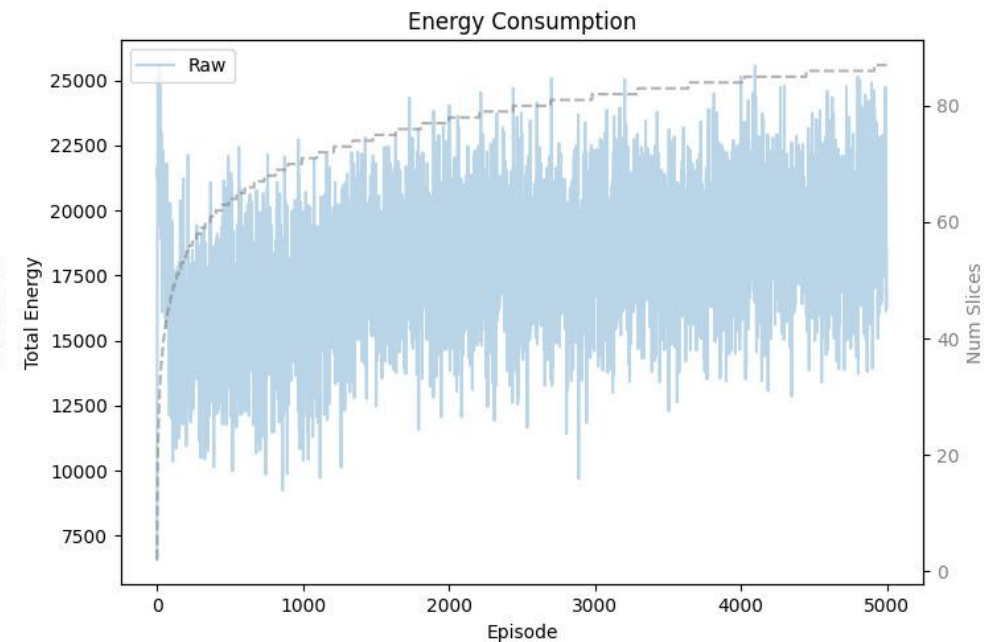
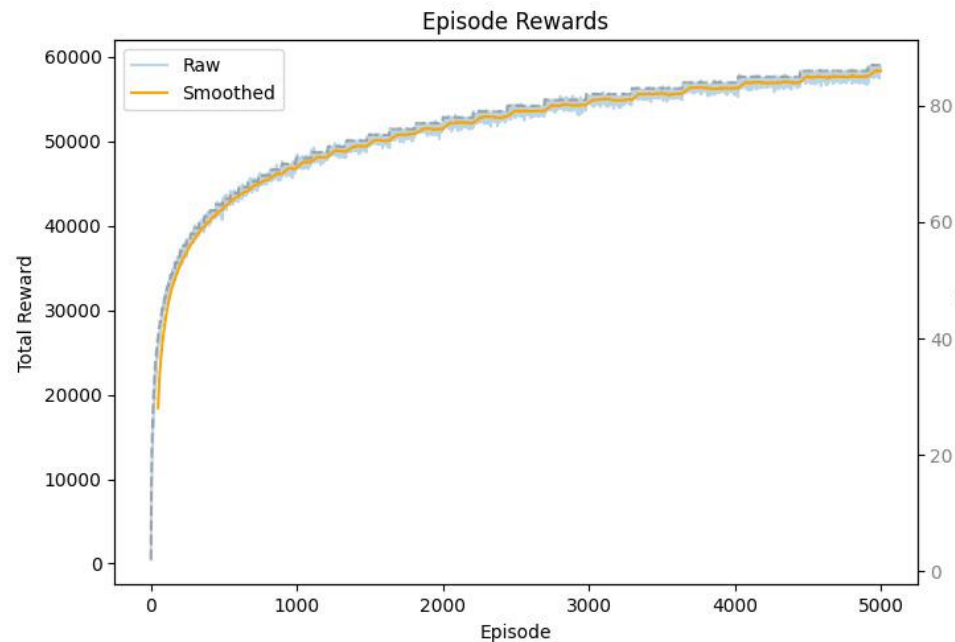
Markov Decision Process
&
Deep Q-Learning Agent



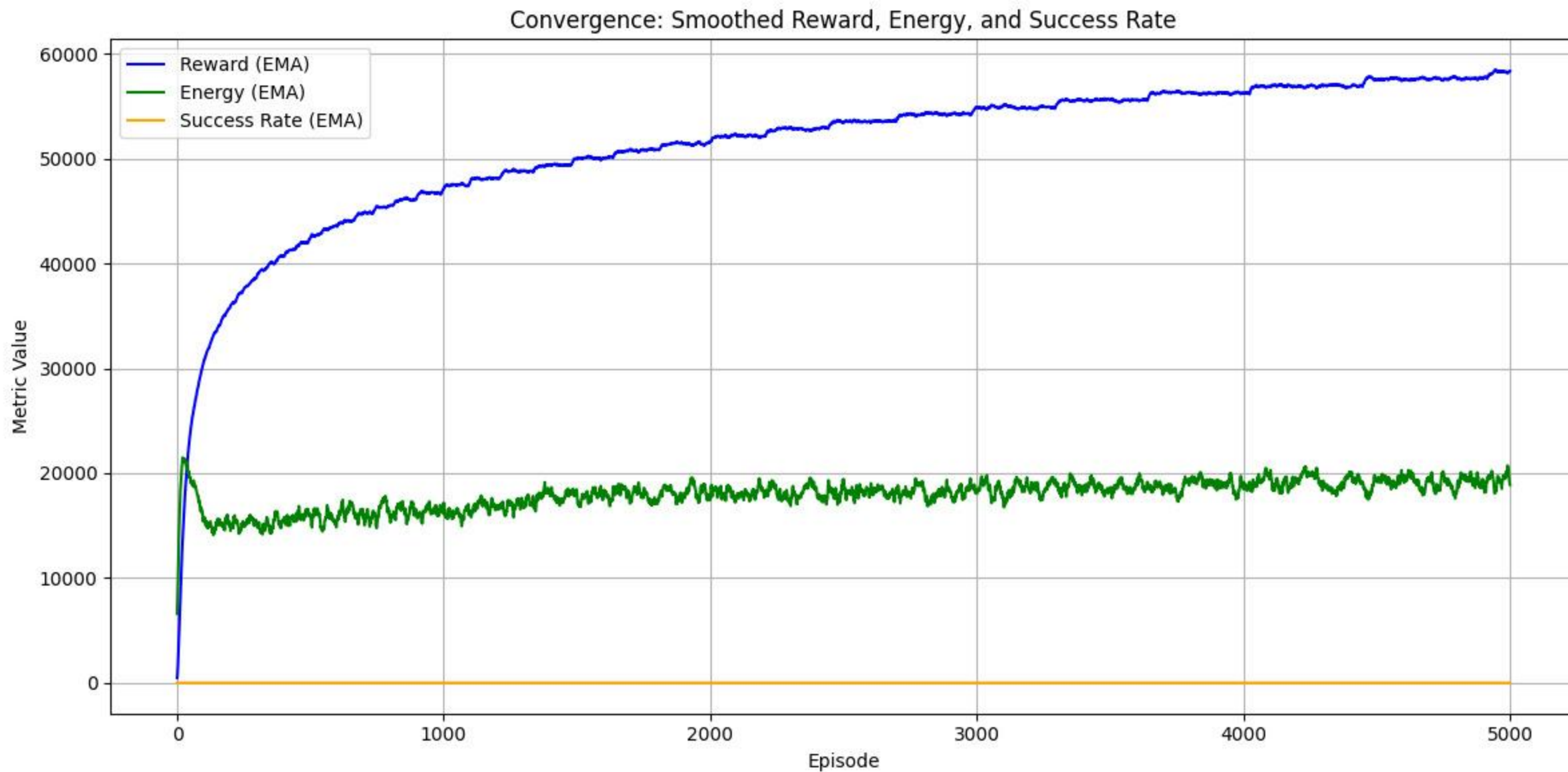
Reward Function:

$$\begin{cases} +\alpha \cdot (\text{QoS_satisfied}) - \beta \cdot \Delta E, \\ -\gamma, \gamma \in [0, 1) \end{cases}$$

Training the DQN Agent

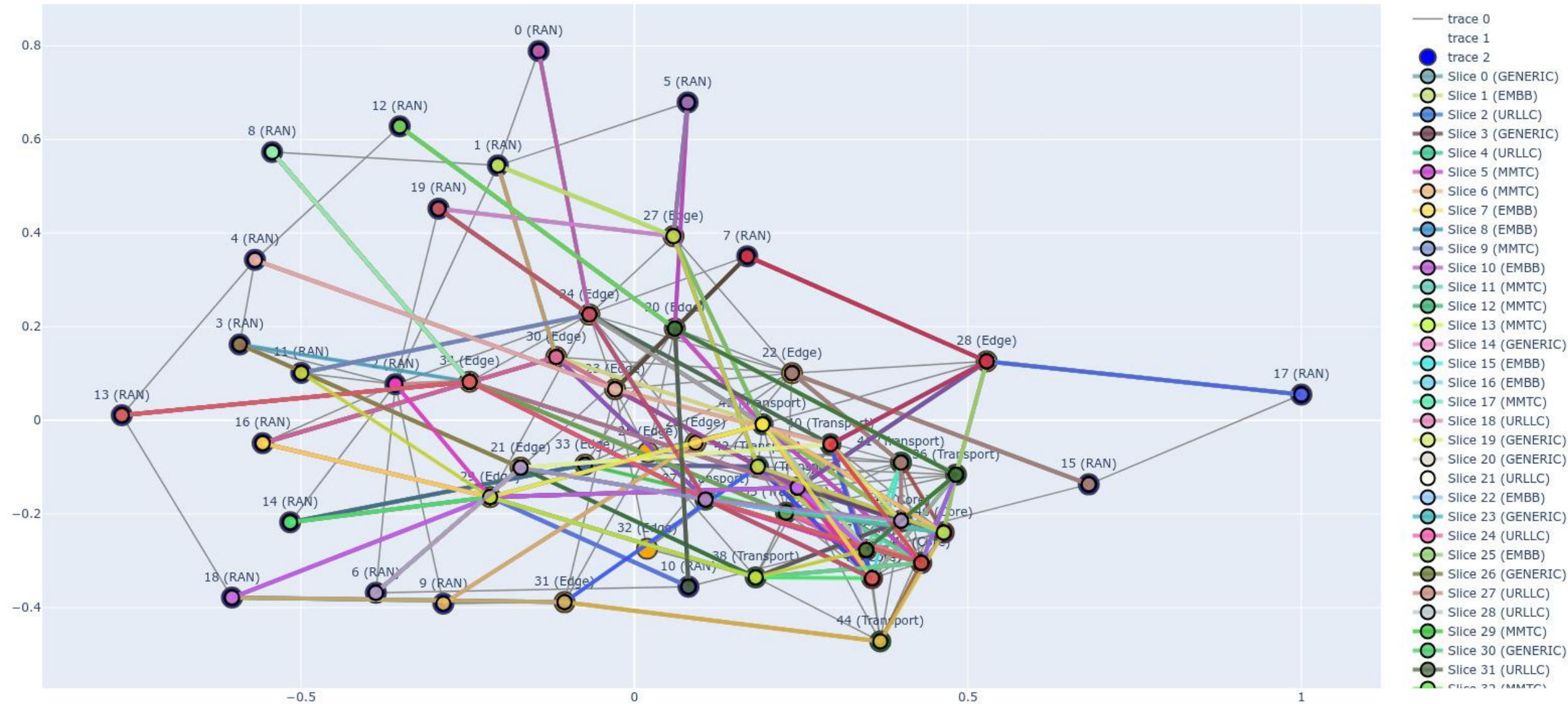


Training the DQN Agent



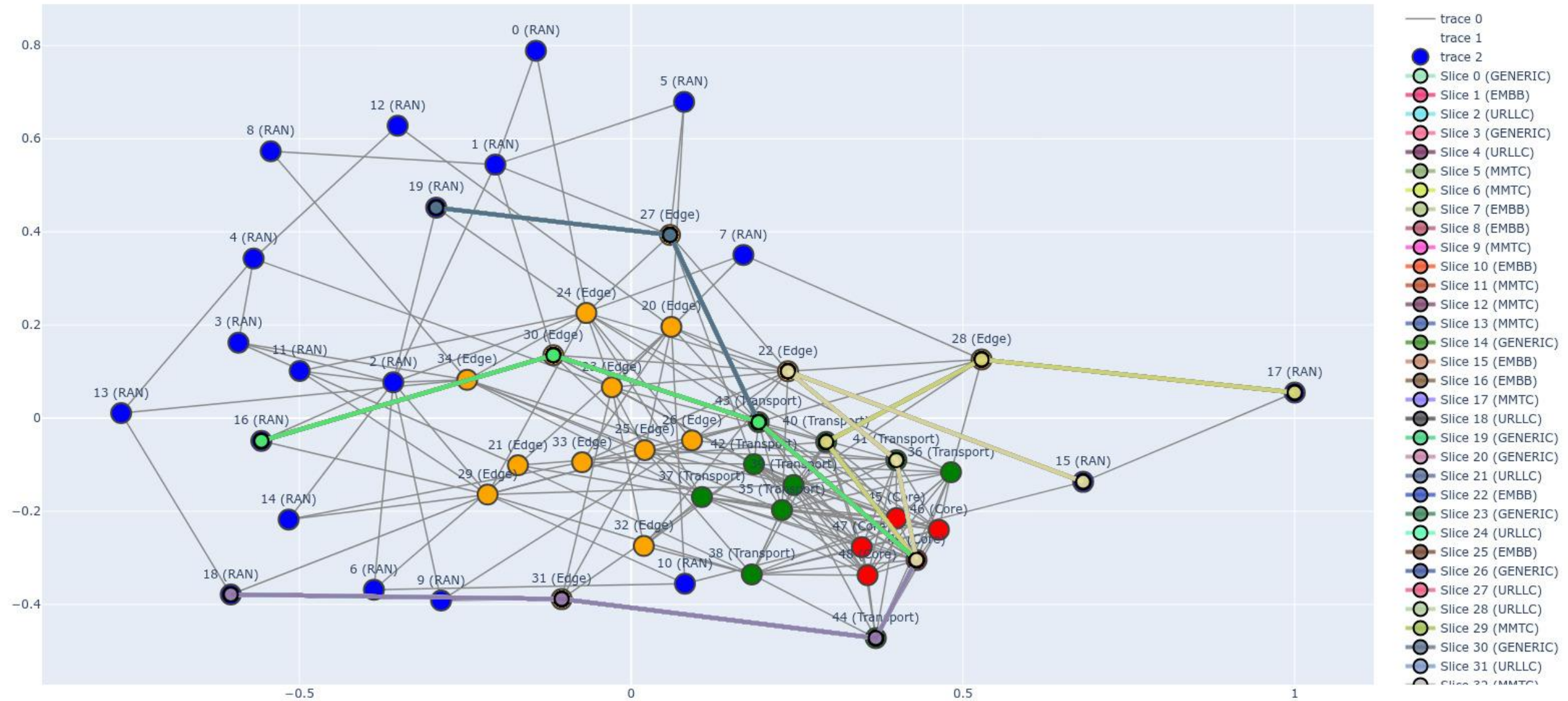
Random Slice Placement

Slice Placement Animation - Slice 149 (GENERIC) - Energy Usage: 35363.465755455974



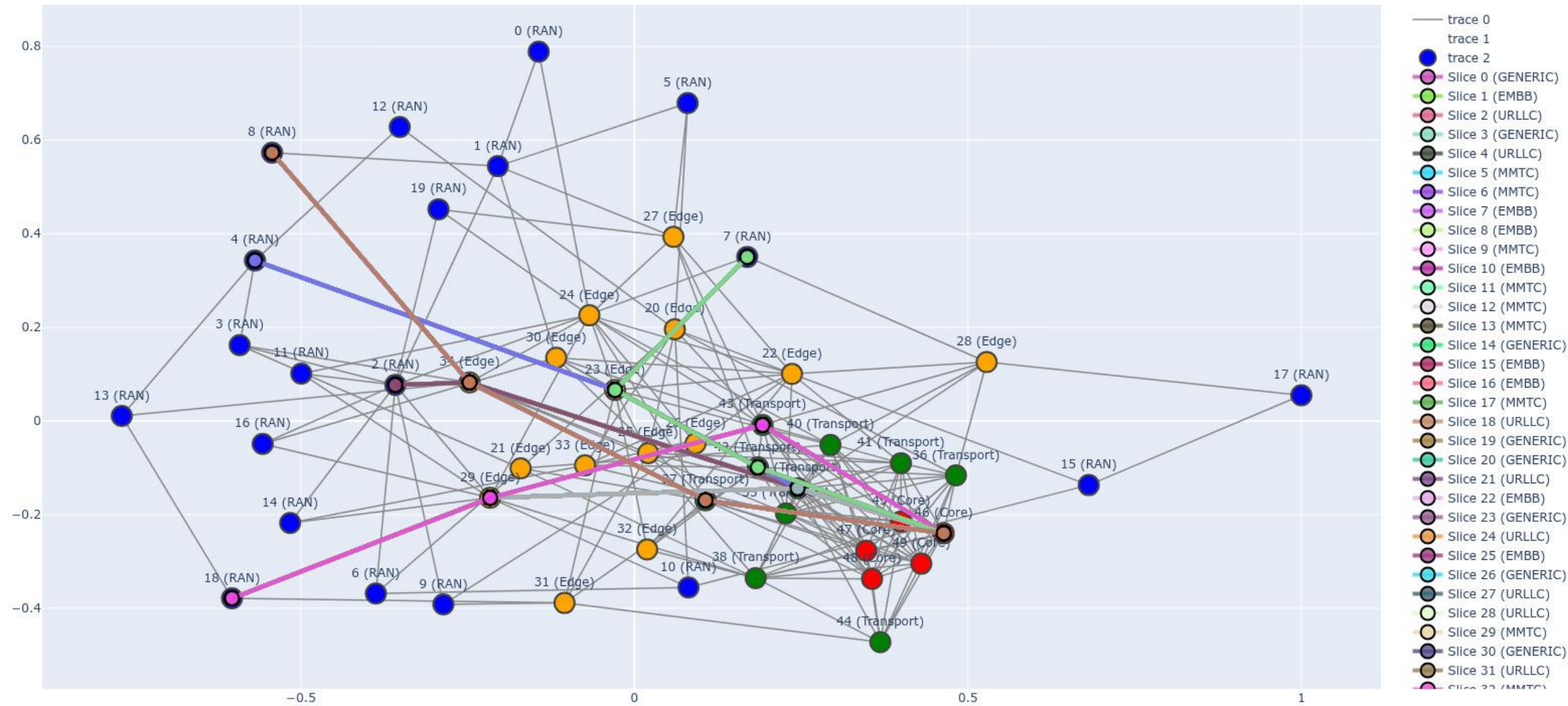
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Slice Placement Animation - Slice 149 (GENERIC) - Energy Usage: 17544.341334375793



DQN Slice Placement

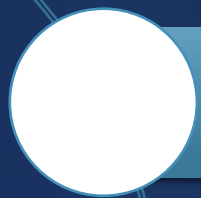
Slice Placement Animation - Slice 149 (GENERIC) - Energy Usage: 16100.893521437638



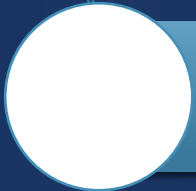
Random vs. Greedy vs. DQN



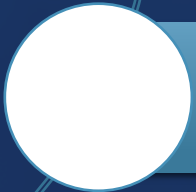
OPEN TO QUESTIONS



Components (topology, agent, environment)



Training Strategies (following up, displays)



Current Limitations & Future Directions



THANK YOU!

ADNEI.DONATTI@USP.BR